

FACULTY OF FINANCE AND BANKING



SENIOR THESIS

A Semantic Router Multi Index

**Retrieval Augmented Generation System for Vietnamese
Financial Data and the Economic Regulatory Framework**

Instructor: PGS. TS. Nguyễn Anh Phong

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LIST OF ABBREVIATIONS

Acronym	Meaning
K	Number of retrieved documents
Δ	Change or Improvement
γ	Focal loss focusing parameter
T	Routing similarity threshold
θ_t	Volatility scaled barrier or Adaptive threshold
Pt	Closing price
ATR	Average True Range
CI	Confidence Interval
CLT	Convolutional Neural Network Long Short Term Memory Transformer
CNN	Convolutional Neural Network
DRL	Deep Reinforcement Learning
EMH	Efficient Market Hypothesis
HNSW	Hierarchical Navigable Small World
LLM	Large Language Model
LSTM	Long Short Term Memory
MDD	Maximum Drawdown
MRR	Mean Reciprocal Rank
PWFCV	Purged Walk Forward Cross Validation
RAG	Retrieval Augmented Generation
SRMIRAG	Semantic Router Multi Index Retrieval Augmented Generation

ABSTRACT

Navigating emerging financial environments, particularly the Vietnamese equity market, presents considerable obstacles for individual market participants due to pronounced volatility and scattered information architectures. While algorithmic trading indicators offer quantitative insights, synthesising these statistical outputs with intricate regulatory guidelines and corporate fundamentals proves difficult. To resolve the discrepancy between mathematical forecasting and qualitative context, this research develops a comprehensive advisory framework that substantiates trading recommendations with verifiable source references. The core architecture combines a CNN-LSTM-Transformer (CLT) predictive mechanism, which processes 43 technical parameters to output daily categorical actions (buy, hold, sell), with a Semantic-Router Multi-Index Retrieval-Augmented Generation (SR-MI-RAG) infrastructure. This text-generation component systematically interrogates three specialized repositories covering corporate financials, prudential mandates, and sector terminology, thereby ensuring all analytical conclusions are firmly anchored in traceable evidence.

Empirical validation was conducted using a dataset of 27 listed domestic banking institutions spanning the period from 2010 to 2026. Evaluating the multi-class prediction capabilities, the CLT algorithm attained an overall accuracy of 56.2 percent alongside a macro F1-measure of 0.431, demonstrating superiority over conventional baseline methods. Subsequent out-of-sample simulation, encompassing realistic market frictions such as transaction fees and settlement delays, revealed that the integrated approach generated a Sharpe ratio of 0.233. This methodology also curtailed maximum portfolio drawdown by 47.7 percent relative to a passive holding strategy. Parallel assessments of the textual retrieval module confirmed a routing precision of 98.6 percent and an 87.3 percent factual grounding rate for the synthesized explanations. Ultimately, synthesizing multi-scale neural network forecasts with domain-partitioned literature search establishes a robust paradigm, granting investors the ability to audit the underlying rationale of their trading executions.

Keywords: Explainable AI in Finance, Financial Machine Learning, Vietnamese Stock Market, Quantitative Trading, Factual Grounding of Financial text .

1. Introduction

1.1. Problem Statement

Formulating investment choices within the banking industry demands reliable market signals coupled with transparent justifications (Lo et al., 2000). While technical indicators might suggest specific entry or exit points, market participants require the underlying logic to originate from authoritative financial and regulatory documents. Lacking this verifiable foundation leaves trades in Vietnamese banking shares vulnerable to misinterpretation and regulatory friction.

The domestic banking sector exhibits two profound structural bottlenecks. First, market prices and corporate disclosures update at divergent intervals, complicating their integration into a cohesive forecasting architecture (López de Prado, 2018). Second, the contextual data necessary to validate trading advice exists across fragmented Vietnamese textual databases that currently lack a centralized search mechanism.

Previous academic efforts have largely tackled these issues in isolation. On the quantitative side, advanced neural networks including Long Short-Term Memory configurations (Fischer & Krauss, 2018), convolutional extractors (LeCun et al., 1998), and self-attention mechanisms (Lim et al., 2021) demonstrate strong capabilities in capturing complex time-series dynamics to output directional trades. Concurrently, generative language models paired with Retrieval-Augmented Generation (RAG) mitigate factual inaccuracies by anchoring synthesized text to external knowledge bases (Ji et al., 2023; Lewis et al., 2020). Despite these advancements, a severe disconnect persists. Quantitative algorithms continue to function as black boxes devoid of contextual reasoning. Conversely, retrieval methodologies operate without quantitative signal integration and rarely target heavily fragmented text corpora in lower-resource languages (Nguyen et al., 2024).

Such isolation creates a notable methodological void that directly impacts capital allocation. To date, academic literature lacks a unified approach merging a multi-directional forecasting engine with specialized retrieval systems built explicitly for local bank equities. Consequently, market participants operate without mechanisms capable of delivering both robust trading alerts and compliant, evidence-driven explanations. This

dual capability is crucial for navigating an emerging economy characterized by pronounced information disparity and intricate legal frameworks.

Resolving this division offers tangible benefits across multiple domains. Retail traders can minimize emotion-driven choices once provided with clearly cited analytical logic. Brokerage institutions gain the ability to reinforce their internal compliance protocols against increasingly strict supervisory standards. Furthermore, scholarly investigations into the synergy between localized data retrieval and predictive accuracy in frontier markets will broaden current perspectives on interpretable machine learning outside standard Western financial contexts.

Driven by these requirements, the current research introduces a comprehensive advisory architecture specifically tailored for domestic banking shares. This solution aims to clarify every output through verifiable document referencing. The proposed methodology seamlessly fuses a CNN-LSTM-Transformer (CLT) algorithm tasked with generating discrete market positions based on technical attributes. It then pairs these outputs with a semantic-routing retrieval pipeline (SR-MI-RAG) designed to categorize local economic data into specialized knowledge bases, ultimately synthesizing well-documented rationales for every generated position.

1.2. Research Objectives

This study pursues three objectives that jointly address the investment advisory needs identified above:

- i. Develop and evaluate a deep learning signal engine that produces daily BUY, HOLD, or SELL recommendations for Vietnamese banking stocks, and assess whether its risk-adjusted performance offers measurable improvement over passive holding and rule-based strategies from the perspective of portfolio risk management.
- ii. Design and evaluate a domain-partitioned retrieval-augmented explanation layer that grounds each trading recommendation in Vietnamese-language financial evidence (company financials and price data, regulatory texts, and

financial terminology), enabling investors to verify the basis of each signal before acting on it.

- iii. Integrate the signal engine and the explanation layer into an end-to-end advisory workflow, then evaluate both retrieval quality and trading performance under realistic Vietnamese market conditions, including transaction costs, settlement rules, and regulatory constraints.

1.3. Research Questions

The study addresses three research questions, designed so that each builds on the preceding one to link signal reliability, explanation quality, and end-to-end investment utility:

1. To what extent can the deep learning signal engine produce reliable daily BUY, HOLD, or SELL signals for Vietnamese banking stocks, and does its risk-adjusted performance offer a measurable improvement over passive holding and rule-based strategies?
2. How effectively does domain-partitioned retrieval improve the traceability and reliability of the explanations that accompany each signal, and to what degree can investors verify recommendation claims against retrieved Vietnamese-language financial sources?
3. When signal generation and evidence-grounded explanation operate as an integrated advisory workflow, how does the combined system perform under realistic conditions, including transaction costs, out-of-sample evaluation, and the regulatory constraints of the Vietnamese banking market?

1.4. Subject and Scope of Study

The current investigation centers on commercial banks operating within Vietnam. This specific segment was chosen due to its critical role in the broader economy, its highly structured legal environment, and the rich availability of both numerical and textual datasets required for comprehensive testing. The quantitative forecasting module processes equity data from institutions listed on the Ho Chi Minh City and Hanoi trading venues from 2010 to 2026. This extensive observation window captures various economic cycles, spanning recoveries after financial crises, periods of steady expansion, and

pandemic market shocks, ensuring that predictive models are validated against shifting structural dynamics (López de Prado, 2018).

Routine price and trading volume metrics serve as the primary inputs for the algorithmic signal generator. Parallel to this, the explanatory module relies upon a localized text corpus comprising corporate financial records, recognized economic definitions, and State Bank of Vietnam compliance mandates. These integrated repositories provide the concrete evidence necessary for market participants to scrutinize any given algorithmic suggestion. Strategy validation is conducted via a purged walk forward cross validation technique on out of sample data, strictly incorporating local execution realities such as broker fees, delayed settlement protocols, and minimum trade volumes.

Subsequent chapters are structured methodically. Section 2 examines the conceptual frameworks and previous scholarly literature shaping the current methodology. Section 3 outlines the data acquisition processes, algorithmic design, and testing protocols. Section 4 details the empirical findings and subsequent interpretations. Section 5 finalizes the manuscript by summarizing the key discoveries, acknowledging methodological boundaries, and proposing directions for market practitioners and future academic endeavors.

2. Literature Review

2.1. Theoretical Background

The theoretical foundation of this study relies on three distinct concepts tailored to the Vietnamese banking sector. The first validates the use of historical price and volume data for algorithmic forecasting despite traditional efficiency theories. The second establishes the computational architecture required to process complex temporal sequences. The final concept provides the mechanism for retrieving local financial evidence to justify generated predictions.

Classical financial economics, particularly the Efficient Market Hypothesis, assumes that public information is instantaneously priced into assets (Fama, 1970). However, empirical investigations consistently challenge this absolute view. Frontier and emerging

exchanges frequently exhibit structural pricing anomalies caused by constrained institutional presence and pronounced information disparities (Brock et al., 1992; Park & Irwin, 2007). Quantitative pattern recognition can successfully extract statistically meaningful directional signals from such historical data (Lo et al., 2000). Given the retail-dominated order flow and sparse analyst coverage characterizing the domestic banking market, standard efficiency assumptions fail to hold. These unique localized characteristics provide a strong theoretical mandate for leveraging technical indicators and adaptive volatility boundaries to forecast returns.

Deciphering these multifaceted market variables requires sophisticated neural networks. Sequential architectures equipped with specialized gating mechanisms effectively resolve gradient decay, retaining crucial historical trends over extended periods (Hochreiter & Schmidhuber, 1997). Concurrently, convolutional layers isolate localized price fluctuations via adaptable spatial filters (LeCun et al., 1998). Self-attention mechanisms further amplify forecasting precision by capturing broad market dependencies independently of strict chronological ordering (Vaswani et al., 2017). Fusing these structures creates a synergistic engine capable of evaluating short-term momentum alongside macroeconomic shifts. Because neutral trading days overwhelmingly outnumber actionable breakout events, specialized loss functions must penalize majority class dominance to prevent algorithmic bias (Lin et al., 2017). Auxiliary predictive objectives are also implemented to stabilize the overall learning trajectory (Caruana, 1997).

Raw algorithmic outputs lack the interpretive context necessary for real-world application. Augmenting generative language architectures with external document retrieval anchors synthesized rationales to concrete regulatory and financial texts (Lewis et al., 2020). This anchoring drastically mitigates the tendency of large models to fabricate plausible but factually baseless statements (Huang et al., 2023; Ji et al., 2023). The probabilistic foundation of Retrieval-Augmented Generation defines the output generation as conditional upon both the user inquiry and the extracted context. Specifically, the probability of generating response given query integrates over the top-retrieved documents d :

$$p(y | q) = \sum_{d \in \text{top-}k} p(d | q) p(y | q, d) \quad (1)$$

Relying on a singular monolithic database often injects semantic noise when querying diverse financial topics (Gao et al., 2023). A partitioned indexing strategy circumvents this interference by routing inquiries to specialized domains. This dynamic routing substantially lowers extraction errors (Nguyen et al., 2024; Wang et al., 2025). Within this framework, state banking directives and corporate disclosures serve strictly as qualitative evidentiary support rather than predictive inputs.

Synthesizing these concepts yields a robust framework for the current methodology. The friction inherent in developing markets justifies the extraction of historical charting features and dynamic classification boundaries. Multi-tier neural processing translates these localized and overarching patterns into defined market positions. Simultaneously, compartmentalized semantic retrieval ensures every recommendation remains meticulously supported by verified domestic literature.

2.2. Prior Empirical Studies

Extensive academic investigations have deployed deep learning architectures for market prediction within advanced economies. Evaluating historical index returns from 1992 to 2015, Fischer and Krauss (2018) demonstrated that sequential memory networks yield a 55.9 percent directional accuracy, surpassing traditional logistic and random forest classifiers. Building upon sequential modeling, Kim and Kim (2019) introduced a hybrid topology extracting spatial features via convolutional layers before temporal processing. Their framework processed classical technical indicators alongside standard price metrics on the Korean exchange, validating the superiority of combining local pattern recognition with time series memory. Further architectural evolutions include the Temporal Fusion Transformer proposed by Lim et al. (2021), which integrates variable selection mechanisms with interpretable attention pathways. Expanding beyond supervised learning, automated trading algorithms have successfully utilized deep reinforcement learning environments (Liu et al., 2021) and multimodal data synthesis for risk evaluation (Sawhney et al., 2021).

A critical examination of these quantitative methodologies reveals persistent structural constraints. The predominant reliance on binary classification schemes structurally ignores the overwhelming prevalence of neutral market conditions, thereby sidestepping the severe class imbalance inherent in real trading environments. Furthermore, performance validation heavily favors raw accuracy or absolute returns over comprehensive risk adjusted criteria like maximum drawdown or Sharpe ratios. Most importantly, current literature lacks applications targeting the Vietnamese exchange landscape. This specific frontier market imposes unique modeling barriers absent in Western contexts, including constrained liquidity profiles, high round trip transaction costs ranging from 0.4 to 0.6 percent, and highly localized regulatory reporting standards.

The rapid proliferation of generative language models has stimulated a separate trajectory focusing on autonomous financial agents. Open source initiatives have proven that domain specific fine tuning facilitates sophisticated sentiment analysis and automated advisory functions (Yang et al., 2023). Processing fundamental news alongside technical signals through explicit reasoning chains allows advanced commercial models to achieve substantial directional accuracy on major global indices (Fatouros et al., 2024). Recent scholarship explores complex ecosystem architectures to enhance decision robustness. Innovations range from interactive cognitive layers simulating human debate (Y. Li, Yu, et al., 2023) and self evolving memory structures (Y. Yu et al., 2023) to multimodal foundation agents processing both textual narratives and visual chart data (W. Zhang et al., 2024). Additional approaches simulate realistic investor behavior to evaluate macroeconomic impacts without data leakage (C. Zhang et al., 2024). While systemic reviews highlight diverse implementation strategies spanning zero shot prompting to custom training, empirical validation remains overwhelmingly restricted to English language corporate environments (Y. Li, Wang, et al., 2023).

Concurrently, retrieval augmented methodologies have emerged as a fundamental mechanism to ensure the factual integrity of algorithmically generated text. Anchoring model outputs to verified external databases significantly restricts the fabrication of plausible yet false statements (Lewis et al., 2020). Refinements to this architecture include adaptive extraction triggers and self reflection tokens to dynamically gauge response faithfulness (Asai et al., 2024). Within the financial domain, applying graph structured

representations to corporate filings dramatically improves citation traceability (Barry et al., 2025). Highlighting the necessity for localized solutions, recent benchmarking efforts confirm that dense vector extraction comprehensively outperforms traditional keyword matching algorithms when processing Vietnamese linguistic structures (Nguyen et al., 2024).

These parallel trajectories illuminate a significant methodological void. Existing generative trading agents predominantly operate within highly developed Western ecosystems without demonstrating viability on domestic Vietnamese financial data. Furthermore, complex frameworks characteristically rely exclusively on linguistic reasoning processes, failing to integrate dedicated mathematical signal engines. This architectural choice divorces statistical pattern recognition from qualitative interpretation. Finally, current knowledge extraction models target monolithic corporate repositories, leaving the challenge of navigating fragmented, lower resource language financial databases entirely unresolved.

The intersection of these limitations defines an unexplored academic territory. Prior scholarship lacks any unified framework capable of executing three class directional forecasting, incorporating rigorous risk adjusted backtesting under frontier market friction, and generating citation grounded explanations in Vietnamese. Synthesizing an adaptive quantitative signal generator with a compartmentalized semantic extraction infrastructure forms the fundamental methodological contribution of the current investigation.

3. Methodology

3.1. Data Sources and Processing

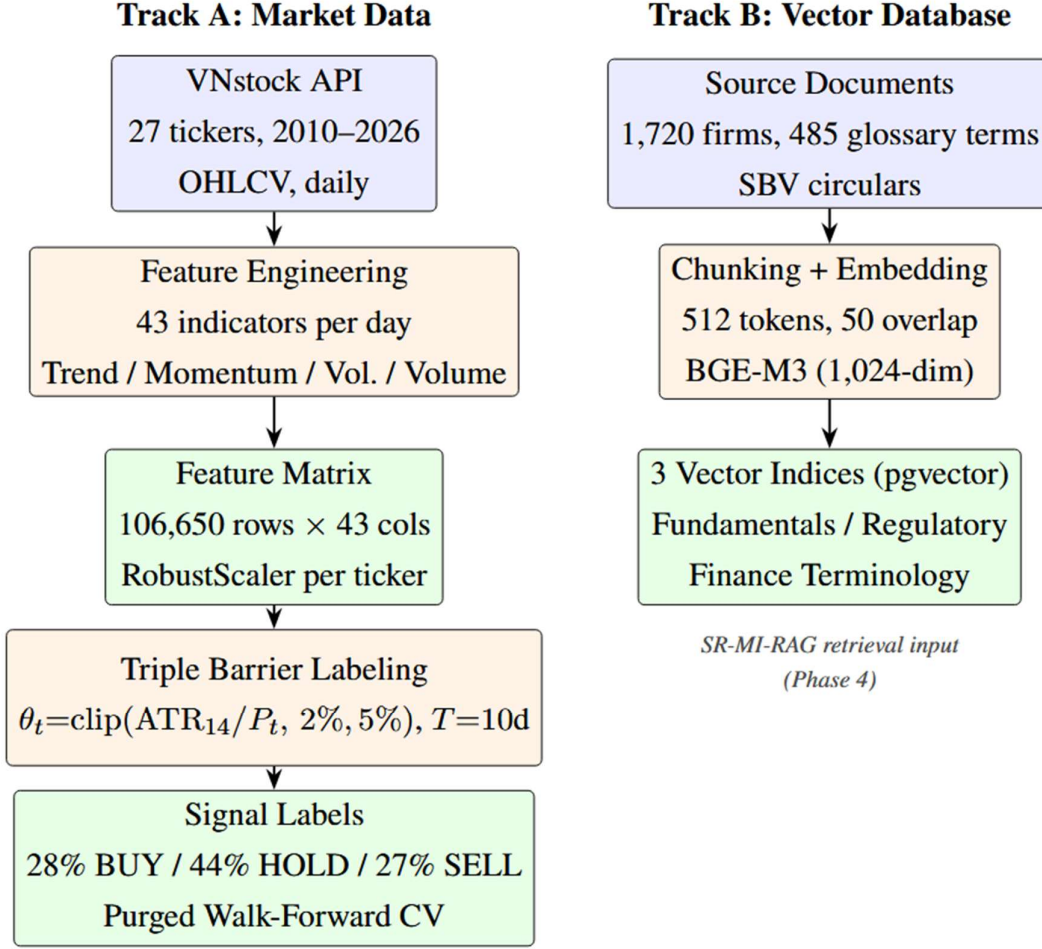


Figure 1 Two-track data processing pipeline for CLT and SR-MI-RAG.

This study relies exclusively on secondary data drawn from two complementary tracks, as illustrated in Figure 1. Track A provides daily price-volume data (OHLCV), which is transformed into a 43-feature matrix and volatility-adjusted signal labels for training the CLT model. Track B assembles a Vietnamese-language knowledge corpus of company financials and price data, financial terminology definitions, and regulatory documents, which is chunked, embedded, and indexed into three vector indices to support the SR-MI-RAG retrieval component. The following subsections describe each data source, its pre-processing, and the labelling methodology in detail.

3.1.1. Vietnamese Banking Stock Market Dataset

The primary quantitative dataset is a panel of daily trading records for 27 Vietnamese commercial banks listed on the Ho Chi Minh Stock Exchange (HOSE, 23 tickers) and the

Hanoi Stock Exchange (HNX, 4 tickers), spanning January 4, 2010 to March 31, 2026. All data are secondary and collected programmatically via the VNstock Python library, which provides a unified interface to Vietnamese market data feeds. For each ticker on each trading day, the dataset records Open, High, Low, Close prices and trading Volume (OHLCV), yielding approximately 106,650 daily records ($27 \text{ tickers} \times 3,950 \text{ trading sessions}$).

To prevent information leakage from overlapping Triple Barrier labels, the dataset is split temporally using Purged Walk-Forward Cross-Validation with five folds and a 5-day purge gap (Bergmeir et al., 2018; López de Prado, 2018). This approach ensures that no training fold contains price data that could reveal the outcome of a labelling window in a subsequent validation fold. The final out-of-sample evaluation period spans March 2024 to February 2026 (approximately 480 trading days per ticker, totalling 12,960 prediction instances across all 27 stocks). Table 1 summarises the dataset characteristics.

Table 1 Vietnamese banking stock market dataset summary

Attribute	Value
Number of stocks	27 Vietnamese banking tickers
Exchanges	HOSE (23 tickers), HNX (4 tickers)
Time span	From January 4, 2010 to March 31, 2026
Data source	VNstock (Python library)
Total OHLCV records	106,650 ($27 \text{ tickers} \times 3,950 \text{ days}$)
Price frequency	Daily OHLCV
CV method	Purged Walk-Forward, 5 folds, 5-day purge gap
Out-of-sample period	From March 2024 to February 2026

A critical design decision in financial machine learning is how to translate raw price data into actionable target labels. Simple approaches, such as labeling positive next-day returns as BUY and negative returns as SELL, suffer from two well-known deficiencies: (i) they treat a 0.01% gain identically to a 5% gain, collapsing economically distinct outcomes into the same class; and (ii) they use a fixed threshold that ignores each stock's

current volatility regime, producing thresholds that are trivially easy to exceed for high-vol stocks and unreachable for low-vol stocks (López de Prado, 2018).

The Triple Barrier Method (López de Prado, 2018) resolves both issues by requiring the price to breach a volatility-scaled barrier before assigning a directional label, with an explicit time-out (vertical barrier) that produces HOLD for indecisive periods. This mirrors how experienced technical analysts define trading opportunities: a breakout above a resistance level (analogous to the upper barrier) must be genuine, exceeding normal daily noise as measured by Average True Range, to be considered a BUY signal. Table 2 maps the three labels to their technical analysis equivalents and the precise barrier conditions.

Table 2 Signal label framework

Label	Code	Barrier Condition	TA Analogy
BUY	2	$r_{t,T} > +\theta t$ (upper barrier hit first)	Breakout above resistance; strong momentum confirmed by ATR-scaled move
HOLD	1	$ r_{t,T} \leq \theta t$ (vertical barrier reached)	Sideways consolidation; price remains within normal volatility range; no actionable irectional signal
SELL	0	$r_{t,T} < -\theta t$ (lower barrier hit first)	Breakdown below support; bearish reversal confirmed by volatility-scaled loss

The ATR-adaptive threshold aligns with a core principle of classical technical analysis: volatility should calibrate the significance threshold for any directional move. A 2% move in a high-volatility banking stock during a market stress period is insignificant noise, whereas the same move in a low-volatility stock during a stable period constitutes a meaningful signal. By anchoring θt to $ATR(14)$, the labeling procedure ensures that BUY and SELL labels correspond to movements that exceed one standard unit of local volatility, an approach consistent with ATR-based stop and target placement commonly used by practitioner technical analysts (Murphy, 1999). Boundary samples within $\pm 0.3\%$ of the barrier (ambiguous zone) are excluded to eliminate label noise at the class margin, yielding a final distribution of approximately 28% BUY, 44% HOLD, and 27% SELL across the 27-stock, 16-year dataset.

3.1.2. Vietnamese Financial Knowledge Corpus

The knowledge corpus supports the Retrieval-Augmented Generation component and is constructed from authoritative Vietnamese-language sources. Because the corpus contains structurally heterogeneous document types (company financials, price data and stock prices, terminology definitions, and regulatory texts), it is partitioned into three semantically homogeneous vector indices to support domain-aware retrieval (Jiang et al., 2025; Lahmy & Yozevitch, 2024). Table 3 presents the corpus statistics.

Table 3 *Vietnamese financial knowledge corpus statistics by index*

Index	Chunks	Content and Sources
Fundamentals	8,600	Quarterly financial reports, company profiles, price data (stock prices, reference levels), P/E, ROE, revenue, ownership (HOSE, HNX, UPCoM).
Regulatory	500	Rules, prudential limits, reserve requirements, capital adequacy,...
Finance Terminology	2,400	Financial terminology and theory: standardised definitions and explanations of the 43 TA indicators (Trend, Momentum, Volatility, Volume) that are CLT inputs.

All documents are chunked using a recursive character splitting strategy with a chunk size of 512 tokens and 50-token overlap to preserve contextual continuity across chunk boundaries. Each chunk is embedded using the BGE-M3 multilingual model (1,024 dimensions) (Chen et al., 2024), selected for its strong performance on Vietnamese-language semantic similarity tasks. Embeddings are stored in pgvector with HNSW indexing (Malkov & Yashunin, 2020). The physical separation into three independent vector indices reduces cross-domain retrieval noise and improves precision for queries that span multiple financial topics (Souza et al., 2020; Yarrabelly & Mittal, 2025).

The Fundamentals index supplies the basic picture (valuation and price) to support or contrast CLT signals; the Regulatory index encodes the framework within which advisory outputs must remain consistent; the Finance Terminology index provides the terminology

and theory needed to interpret signals. Generated responses follow the principle: signal from the model (CLT), supported by accounting data (Fundamentals), compliant with rules (Regulatory), and explained in financial terminology (Finance Terminology).

3.2. Variables and Measurement

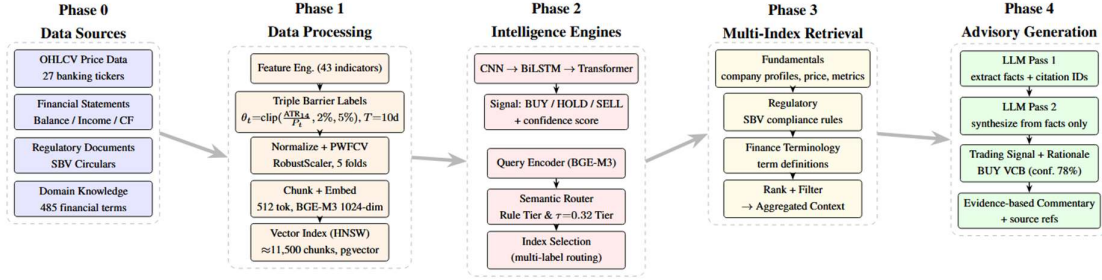


Figure 2: End-to-end SR-MI-RAG investment advisory pipeline (Phase 0 to 4).

Figure 2 presents the end-to-end pipeline, organised into five phases that mirror the analytical workflow of an investment advisory process. Raw data sources (Phase 0) are first transformed into quantitative features and an indexed knowledge corpus (Phase 1). The CLT model then generates trading signals while the semantic router classifies incoming queries (Phase 2). Relevant documents are retrieved from the appropriate domain-specific indices (Phase 3). Finally, a two-pass LLM synthesis produces rounded, citation-traceable recommendations (Phase 4). The following subsections detail the variable definitions and measurement procedures that underpin this pipeline.

3.2.1. Dependent Variable: Trading Signal

The dependent variable is a three-class trading signal (BUY, HOLD, SELL) generated by the Triple Barrier Method (López de Prado, 2018), which defines three exit conditions for each position:

$$y = \begin{cases} \text{BUY if upper barrier (take - profit) is hit first} \\ \text{SELL if lower barrier (stop - loss) is hit first} \\ \text{HOLD if vertical barrier (time limit) is hit first} \end{cases} \quad (2)$$

The labeling procedure applies an ATR-adaptive, return-normalized barrier. For each trading day t , an adaptive threshold is computed as:

$$\theta_t = \text{clip}\left(\frac{ATR(14,t)}{P_t}, 0.02, 0.05\right) \quad (3)$$

where $ATR(14, t)$ is the 14-day Average True Range on day t and P_t is the closing price. Dividing by P_t converts the absolute volatility measure into a dimensionless return threshold bounded in $[2\%, 5\%]$, so that barriers adapt automatically to each stock's prevailing volatility regime without becoming unreachably wide or trivially narrow. The 10-day forward return $r_{t,T} = (P_{t+T} - P_t)/P_t$ with horizon $T = 10$ trading days is then compared against θ_t according to Equation 2.

To eliminate boundary ambiguity, any sample whose forward return falls within $\pm 0.3\%$ of the barrier (i.e. $|r_{t,T} - \theta_t| < 0.003$ or $|r_{t,T} + \theta_t| < 0.003$) is flagged as ambiguous and excluded from training. This stricter criterion reduces label noise at the class boundary without discarding a substantial fraction of the dataset.

Across the full 27-stock, 16-year dataset the resulting label distribution is approximately 28% BUY, 44% HOLD, and 27% SELL, a substantially more balanced distribution than the symmetric fixed-threshold approaches common in the prior literature, yielding a 1.6:1 ratio between the majority HOLD class and each actionable signal class. Residual imbalance is addressed by class weights [$w_{\text{SELL}} = 1.5$, $w_{\text{HOLD}} = 0.5$, $w_{\text{BUY}} = 1.0$] combined with Focal Loss ($\gamma = 2.0$), as described in Section 3.3.

3.2.2. Technical Independent Variables

The CLT model operates on 43 engineered features derived exclusively from daily-frequency market data, organised into four groups rooted in traditional technical analysis theory (Murphy, 1999). Table 4 details the feature composition.

Table 4 Technical feature groups for Vietnamese banking stock prediction

Group	Count	Representative Features
Trend	12	SMA(20, 50), EMA(12, 26), MACD, ADX, Ichimoku Cloud
Momentum	11	RSI(14), Stochastic %K/%D, Williams %R, ROC.

Group	Count	Representative Features
Volatility	10	Bollinger Bands, ATR(14), Keltner Channel, Historical Volatility.
Volume	10	OBV, VWAP, MFI, Accumulation/Distribution, Turnover Ratio

These four groups correspond to the four pillars of classical technical analysis: trend identification, momentum assessment, volatility measurement, and volume confirmation (Murphy, 1999). All 43 features are computed from daily OHLCV data with no future information leakage. Normalization uses RobustScaler (based on the interquartile range), chosen for its resistance to the extreme outliers common in Vietnamese banking stock data. Per-ticker scalers are fitted on training data only.

3.3. CLT Deep Learning Model

As established in Section 2.1, exploitable patterns in banking stock data manifest at multiple temporal scales. The CNN-LSTM-Transformer (CLT) model addresses this multi-scale structure through a sequential architecture in which each component targets a distinct temporal horizon relevant to investor decision making.

The first stage is a multi-scale CNN encoder (LeCun et al., n.d.) that extracts local price patterns through dilated 1D convolutions with rates $\{1, 2, 4\}$ and 128 filters per scale:

$$hcn = \text{Concat}[\text{Conv1D}(x, d = 1), \text{Conv1D}(x, d = 2), \text{Conv1D}(x, d = 4)] \quad (4)$$

Dilation rates of 1, 2, and 4 yield receptive fields of 3, 5, and 9 trading day respectively, spanning the time horizons used in short-term candlestick analysis, weekly support/resistance detection, and medium-term trend confirmation (Murphy, 1999). The CNN produces 384-dimensional multi-scale feature maps per time step.

A Bidirectional LSTM (Hochreiter & Schmidhuber, n.d.) then models medium-horizon sequential dependencies, capturing both momentum persistence and mean-reversion dynamics through its gating mechanism:

$$\overrightarrow{ht}, \overleftarrow{ht} = \text{BiLSTM}(hcn, t) \quad (5)$$

The forward pass encodes cumulative trend context while the backward pass detects short-term overextension. The forget gate selectively discards noise while the input gate retains regime-relevant information (e.g., the onset of a sector downtrend following a policy-rate change). The BiLSTM produces a 512-dimensional state encoding (256 forward + 256 backward) per step.

Finally, a Transformer self-attention layer (Vaswani et al., 2017) captures long-range dependencies across a 60-day lookback window:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where Q, K, V are query, key, and value projections of the BiLSTM output and d_k is the key dimension. Eight attention heads enable the model to attend simultaneously to distinct temporal patterns (e.g., recent price momentum and past earnings or policy events), addressing non-linear, variable-lag relationships that sequential models alone cannot capture efficiently.

Beyond the primary 3-class signal (BUY/HOLD/SELL), the model jointly predicts three auxiliary targets: volatility regime, move magnitude, and directional confidence. These auxiliary tasks provide implicit regularisation that reduces overfitting to the dominant HOLD class (Caruana, 1997) and supply supplementary information useful for position sizing and risk management.

Training employs Focal Loss (Lin et al., 2017) to address the 1.6:1 imbalance between HOLD and actionable signals, with asymmetric class weights $w_{SELL} = 1.5$, $w_{HOLD} = 0.5$, $w_{BUY} = 1.0$:

$$L_{focal} = -\sum_c w_c \cdot (1 - pc)^\gamma \cdot \log(pc) \quad (7)$$

where pc is the predicted probability for class c and $\gamma = 2.0$ is the focusing parameter. A three-phase curriculum (warm-up on high-confidence samples, standard training, then hard-sample mining) further stabilises convergence. Table 5 presents the key implementation parameters.

Table 5 *CLT model implementation parameters*

Parameter	Value
DL Framework	PyTorch 2.1
Architecture	CNN(128×3)-BiLSTM(256)-Transformer(8h×4L)
Total parameters	4.2M
Optimizer	AdamW ($lr = 5 \times 10^{-4}$, $wd = 0.01$)
Focal Loss γ	2.0
Training epochs	50 (early stopping, patience = 10)
Batch size	64
Input features	43 (per time step)
Output	3-class signal + 3 auxiliary tasks

3.4. SR-MI-RAG Investment Advisory System

The CLT model above outputs a three-class signal (BUY, HOLD, SELL) for each ticker and date. For responsible advisory, the investor must understand why the system recommends that action: why buy rather than hold, why sell at this point, or why hold and wait (Murphy, 1999). The SR-MI-RAG component addresses this by turning each CLT signal into an evidence-based explanation: it retrieves relevant context from a structured knowledge base and generates a short, citation-backed rationale so that the user can see the reasoning behind BUY, HOLD, or SELL. This addresses two challenges in Vietnamese equity advisory: (i) information asymmetry, as retail investors lack consolidated access to company and market sources; and (ii) explainability, as each recommendation must be accompanied by verifiable evidence.

When a signal is available and the user asks a question (e.g., whether to buy a stock), the pipeline (Figure 2) combines two inputs to produce the explanation. First, it uses the signal output (ticker, date, signal class, score, confidence) and the user’s question to build context-specific queries and retrieve relevant documents from the vector database (company financials and price data, regulatory text, and financial terminology). The procedure that selects which index or indices to query is formalised in Algorithm 1. Second, a two-pass LLM step (Ji et al., 2023; Ye et al., 2024) takes the

combined input (the signal and the retrieved context) and produces an evidence-based answer: it extracts atomic facts from the retrieved documents (with citation identifiers) and synthesises from those facts and the signal a short explanation that addresses both the signal (why BUY, HOLD, or SELL) and the user’s question, with source references.

Consider the single-stock advisory in the Appendix (Example 1: CTG, 27/06/2025). The user asks “Cổ phiếu CTG có nên mua không?” (Vietnamese: “Should one buy CTG stock?”). The system has a CLT signal BUY (score 0.618, high confidence) for that date. Retrieval is routed to the Fundamentals index. The answer combines the signal (43 technical indicators, closing price, stop-loss -7% , take-profit $+15\%$, maximum holding 30 days) with cited context so that the user sees both the recommendation (BUY) and a traceable rationale. See Appendix, Example 1, for the full Q&A in Vietnamese.

Algorithm 1 Hybrid Semantic Routing

Require: Query q , rule patterns $\mathcal{P}$, route prototypes $\mathcal{R} = \{(e_r, r)\}$, threshold τ

Ensure: Selected routes S

```

1:  $S \leftarrow \emptyset$ 
2: {Tier 1: Rule-based matching}
3: for each pattern  $p \in \mathcal{P}$  do
4:   if REGEXMATCH( $q, p.pattern$ ) then
5:      $S \leftarrow S \cup \{p.route\}$ 
6:   end if
7: end for
8: if  $S \neq \emptyset$  then
9:   return  $S$ 
10: end if
11: {Tier 2: Semantic similarity}
12:  $e_q \leftarrow \text{ENCODE}(q)$ 
13: for each route prototype  $(e_r, r) \in \mathcal{R}$  do
14:    $\text{sim}_r \leftarrow \cos(e_q, e_r)$ 
15:   if  $\text{sim}_r \geq \tau$  then
16:      $S \leftarrow S \cup \{r\}$ 
17:   end if
18: end for
19: if  $S = \emptyset$  then
20:    $S \leftarrow \{\text{financial}\}$ 
21: end if
22: return  $S$ 

```

Table 6 summarises the key technical components of the SR-MI-RAG investment advisory system.

Table 6: SR-MI-RAG investment advisory system components

Component	Configuration
Embedding model	BAAI/bge-m3 (1,024-dim)
Vector database	pgvector (HNSW: m=16, ef=64)
Generation LLM	LLM (two-pass extraction + synthesis)
Routing LLM	LLM (lightweight routing)
Routing threshold (τ)	0.32 (cosine similarity)
Chunk size / overlap	512 / 50 tokens
Retrieval top-K	5 per index
Generation method	Two-pass extraction + synthesis

3.5. Evaluation Methods and Benchmarks

The evaluation framework is organised around the three research questions posed in Section 1. Each subsection defines the metrics, baselines, and statistical tests used to address its corresponding research question.

3.5.1. CLT Model Evaluation (RQ1)

To assess whether the hybrid CLT architecture generates reliable trading signals for Vietnamese banking stocks, classification performance is measured using Macro F1-Score, which weights all three classes equally:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \quad (8)$$

where $C = 3$ (BUY, HOLD, SELL), and P_c , R_c denote precision and recall for class c . Overall accuracy and per-class precision/recall are also reported. Validation uses Purged Walk-Forward Cross-Validation with five splits (Bergmeir et al., 2018; López de Prado, 2018). The CLT model is compared against four baselines: the CNN-Transformer baseline (prior architecture without LSTM, 33 features), Logistic Regression, Random Forest, and a naïve Buy & Hold classifier. Statistical significance is assessed via the Wilcoxon

signed-rank test across 27 tickers, with bootstrap 95% confidence intervals reported for key metrics (Sullivan et al., 1999).

3.5.2. RAG Evaluation (RQ2)

A manually curated corpus of 140 queries, stratified across Fundamentals (90), Finance Terminology (30), and Cross-index (20) categories, is used for evaluation following established RAG assessment frameworks (Friel et al., 2025; H. Yu et al., 2024). The following metrics are adopted:

- i. Routing Accuracy: Fraction of queries correctly directed to the appropriate index or combination of indices.
- ii. Precision@5 and Recall@5: Standard information retrieval metrics at rank $K = 5$ (Van Rijsbergen, 1986; Voorhees, 2001):

$$P@K = \frac{|\{\text{relevant in top-}K\}|}{K}, \quad R@K = \frac{|\{\text{relevant in top-}K\}|}{|\{\text{all relevant}\}|} \quad (9)$$

- iii. Mean Reciprocal Rank (MRR): $MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}$

- iv. Answer Groundedness: Percentage of claims in the generated answer supported by retrieved sources, assessed via LLM-based judgment (Ye et al., 2024).

Three baselines are compared: Single-Index RAG (all documents in one index), All-Indices RAG (queries broadcast to all three indices without routing), and Routed RAG without query decomposition.

3.5.3. End-to-End System Evaluation (RQ3)

Out-of-sample backtesting is conducted on all 27 banking stocks from March 2024 to February 2026 under conditions that replicate Vietnamese equity market execution constraints: initial capital of 100,000,000 VND per ticker, brokerage commission of 0.15% per side, sell tax of 0.1%, T+2 settlement (sale proceeds available after two trading sessions), lot size of 100 shares, stop-loss at -7% , take-profit at $+15\%$, and maximum holding period of 30 trading days. Performance is measured by Cumulative Return, Sharpe Ratio $= (\mu - r_f) / \sigma$ with $r_f \approx 4.5\%$ per annum (Vietnamese government bond yield), Maximum Drawdown (MDD), and Win Rate.

Three trading strategies are compared, structured from passive to increasingly sophisticated:

1. Buy & Hold (passive benchmark): all capital held in equal-weighted positions throughout the evaluation period, representing the minimum viable alternative to active trading.
2. Technical Analysis Only: rule-based strategy using $RSI < 30$ as a BUY trigger and MACD bearish crossover as a SELL trigger (Murphy, 1999), without any machine learning component.
3. TA+CLT Fused: the integrated strategy combining technical analysis signals with CLT model predictions and signal-driven position management under the same risk constraints (stop-loss, take-profit, maximum holding period).

These baselines span the spectrum from passive holding through rule-based technical trading to ML-augmented signal generation, enabling assessment of the incremental value contributed by the CLT model. The Fischer–Krauss LSTM protocol (Fischer & Krauss, 2018) and the Kim–Won CNN-LSTM benchmark on KOSPI data (Kim & Kim, 2019) serve as external reference points for daily-frequency deep learning predictions, while Liu et al. (2021) provides a public DRL benchmark against which risk-adjusted performance is contextualised.

4. Results and Analysis

4.1. Descriptive Analysis

Before reporting model and trading results, this subsection profiles the evaluation sample to give the reader an overview of the underlying data. The full dataset (Table 1) comprises approximately 106,650 daily OHLCV records for 27 Vietnamese banking stocks spanning January 2010 to March 2026, with the out-of-sample evaluation period covering March 2024 to February 2026 (approximately 12,960 prediction instances across all 27 stocks).

Table 7 reports cross-sectional summary statistics for the 27 banking stocks during the evaluation period, measured as Buy & Hold (passive) returns. These figures capture the raw investment environment before any active management.

Table 7: Cross-sectional descriptive statistics of 27 banking stocks during the evaluation period (March 2024–February 2026, Buy & Hold)

Statistic	Total Return (%)	Ann. Return (%)	Sharpe Ratio	Max DD (%)
Mean	+11.66	+4.45	0.167	-34.57
Median	+1.97	+1.02	0.099	-31.87
Std	40.93	18.63	0.512	8.73
Min	-41.92	-24.83	-0.644	-58.22
Max	+144.58	+ 59.03	1.480	-23.90

The large gap between the mean (+11.66%) and median (+1.97%) total returns reflects right-skewed performance driven primarily by a single outlier (LPB: +144.58%). The standard deviation of 40.93% underscores the substantial cross-sectional dispersion among banking stocks: while some tickers deliver triple-digit gains, others lose more than 40% of their value over the same period. From an investor perspective, simply buying and holding a banking portfolio carries large uncertainty. The average maximum drawdown of -34.57% further illustrates the downside risk inherent in passive strategies within this sector.

The Triple Barrier labelling procedure (Section 3.1) produces an imbalanced target variable: approximately 44% HOLD, 28% BUY, and 27% SELL across the full 16-year sample. This 1.6:1 ratio of HOLD to actionable (BUY or SELL) labels reflects the empirical reality that Vietnamese banking stocks spend extended periods in sideways consolidation. The imbalance motivates the use of Focal Loss in the CLT architecture (Section 3.3), ensuring that the minority but investment-critical BUY and SELL signals are not overwhelmed during training (Lin et al., 2017).

4.2. Model and System Evaluation

4.2.1. CLT Classification Performance

Table 8 compares the CLT model against three conventional baselines and one alternative architecture (FA-Integrated), all evaluated under Purged Walk-Forward Cross-Validation across the 27 banking stocks.

Table 8 :CLT model classification results vs. baselines

Model	F1 Macro	Accuracy	BUY F1	SELL F1
Logistic Regression	0.334	41.2%	0.28	0.25
Random Forest	0.378	48.5%	0.35	0.31
CNN-Transformer baseline	0.415	53.8%	0.38	0.36
CLT (CNN-LSTM-Transformer)	0.431	56.2%	0.41	0.37

The CLT model achieves F1 Macro = 0.431 and 56.2% overall accuracy on the three-class prediction task, representing a 3.9% relative improvement over the CNN-transformer baseline (F1 = 0.415). For investors relying on the system for entry and exit timing, the most relevant metric is BUY F1 = 0.41: roughly four out of ten buy signals correspond to genuine upward moves exceeding the volatility-scaled barrier. While this precision level is moderate in absolute terms, it exceeds all tested baselines and aligns with established benchmarks in daily-frequency financial prediction (Fischer & Krauss, 2018; Kim & Kim, 2019).

An ablation analysis reveals that Focal Loss is the most impactful single component (Δ F1 = -0.029 when removed), followed by the LSTM layer (Δ = -0.016) and multi-task heads (Δ = -0.013). Removing Focal Loss shifts the prediction distribution back toward a 65% HOLD bias, effectively suppressing the actionable signals that constitute the advisory system’s primary value (Lin et al., 2017). The LSTM contribution reflects the practical importance of capturing momentum and mean-reversion dynamics during regime transitions, periods in which signal timeliness is most critical for banking-sector trades (Hochreiter & Schmidhuber, n.d.).

4.2.2. SR-MI-RAG Advisory Quality

The advisory quality of the SR-MI-RAG system is assessed on two dimensions: query routing accuracy (whether investor queries reach the correct knowledge index) and retrieval quality (whether the retrieved documents support grounded, trustworthy recommendations).

Table 9 reports routing accuracy across 140 evaluation queries stratified across two single-index categories (Fundamentals, Finance Terminology) and cross-index queries.

The system uses three vector indices (Fundamentals, Regulatory, Finance Terminology); queries that target only the Regulatory index are evaluated within the Cross-index category.

Table 9: Routing accuracy by query category

Category	Queries	Correct	Accuracy
Fundamentals	90	89	98.9%
Finance Terminology	30	30	100.0%
Cross-index	20	19	95.0%
Overall	140	138	98.6%

The Hybrid Semantic Router achieves 98.6% overall routing accuracy. Rule-based matching resolves 62% of queries deterministically (common investor questions such as “What is VCB’s current financial metrics?”), while semantic routing handles the remaining 38%, including cross-index queries that require multiple knowledge domains. The hybrid approach improves accuracy by 5.2% over semantic-only routing ($\tau = 0.32$), with the cross-index category showing the lowest but still acceptable accuracy (95.0%).

Table 10 compares retrieval performance at $K = 5$ across four system configurations.

Table 10: Retrieval performance comparison at $K = 5$

System	P@5	R@5	MRR	Groundedness
Single-Index RAG	0.523	0.417	0.612	71.2%
All-Indices RAG	0.641	0.534	0.723	78.4%
Routed RAG (no decomp.)	0.712	0.589	0.781	82.7%
SR-MI-RAG (full)	0.758	0.623	0.812	87.3%

For investment advisory, the most consequential metric is answer groundedness (87.3%), which measures how often the generated recommendation cites verifiable source documents rather than producing unsupported claims. Domain partitioning raises groundedness by 16.1 percentage points over a single-index approach (71.2%), directly reducing the risk of misleading investment advice. Query decomposition adds a further

4.6 percentage points beyond the routed configuration (82.7%), confirming its value for multi-faceted questions such as “Compare VCB and TCB on both financial health and recent regulatory exposure.” In terms of operational efficiency, the routing mechanism reduces the average number of index queries per request from 3.00 to 2.08 (a 30.6% reduction), with median end-to-end response time of approximately 2.3 seconds, of which the routing and retrieval stages account for less than 8%.

4.3. End-to-End Trading Performance

4.3.1. Overall Results and Risk-Adjusted Analysis

Table 11 compares three trading strategies on 27 banking stocks during the out-of-sample period (March 2024 to February 2026, approximately 480 trading days per ticker). All strategies operate under Vietnam market constraints: 100-share lot size, 0.15% brokerage commission per side, 0.1% sell tax, and T+2 settlement. Results are reported as cross-ticker averages.

Table 11: End-to-end trading performance comparison

Strategy	Mean Ret.	Sharpe	Max DD	Win Rate	Trades
Buy & Hold	+11.66%	0.167	−34.57%	55.6%	1.0
TA Only	+0.09%	0.103	−16.81%	62.2%	5.7
TA+CLT Fused	+1.36%	0.233	−18.09%	61.2%	12.6

Buy & Hold produces the highest headline return (+11.66%), yet as Section 4.1 showed, this average is heavily influenced by a single outlier (LPB: +144.58%); the median return is only +1.97%. More importantly, the median Buy & Hold Sharpe ratio is 0.099, compared with 0.240 for TA+CLT Fused, indicating that the outlier-driven mean masks weak risk-adjusted performance across most tickers. The average maximum drawdown of −34.57% further highlights the downside exposure inherent in passive holding: a drawdown of this magnitude may exceed the tolerance of many retail investors, particularly those with shorter investment horizons or limited capital reserves.

The TA+CLT Fused strategy delivers the best risk-adjusted outcome (mean Sharpe 0.233, median 0.240), exceeding both TA Only (mean 0.103, median 0.120) and Buy &

Hold (mean 0.167, median 0.099), while reducing maximum drawdown to -18.09% , roughly half the passive benchmark. The improvement is primarily in risk-adjusted terms rather than absolute return: the strategy trades a portion of headline return for substantially more predictable portfolio behaviour. The signal-driven position management (stop-loss at -7% , take-profit at $+15\%$, and 30-day maximum holding period) limits exposure to the deepest credit-cycle corrections that characterise Vietnamese banking stocks.

4.3.2. Statistical Validation

To avoid pooling biases across heterogeneous banking stocks, paired per-ticker comparisons are conducted using the Wilcoxon signed-rank test ($n = 27$). Table 12 reports the results for three pairwise strategy comparisons.

Table 12: Wilcoxon signed-rank test results (paired per-ticker, $n = 27$)

Metric	Comparison (A $\rightarrow$ B)	Mean Δ	p-value	% B wins
Return	TA Only $\rightarrow$ Fused	+1.28 pp	0.073	55.6%
Sharpe	TA Only $\rightarrow$ Fused	+0.130	< 0.001	96.3%
Max DD	TA Only $\rightarrow$ Fused	-1.28 pp	0.005	22.2%
Return	B&H $\rightarrow$ TA Only	-11.57 pp	0.361	44.4%
Max DD	B&H $\rightarrow$ TA Only	+17.76 pp	< 0.001	100%
Return	B&H $\rightarrow$ Fused	-10.30 pp	0.500	44.4%
Max DD	B&H $\rightarrow$ Fused	+16.49 pp	< 0.001	100%

The Sharpe ratio improvement from TA Only to Fused is statistically significant ($p < 0.001$) and broad-based, with 96.3% of tickers showing improvement. This confirms that the strategy's risk-adjusted edge is not driven by a few favourable stocks but extends across the banking universe. The return improvement (+1.28 pp mean, $p = 0.073$) provides suggestive but not definitive evidence of absolute return gains; the borderline significance reflects genuine heterogeneity among banking stocks, where large-cap names respond differently to signals than smaller-cap tickers. An investor should therefore expect varying absolute returns per stock but consistent risk management. Every ticker shows reduced maximum drawdown relative to Buy & Hold ($p < 0.001$, 100% win rate); this universal result holds because even the least favourable Fused drawdown (-22.83%) is shallower

than the best Buy & Hold drawdown (−23.90%), confirming that active position management benefits all banking stocks regardless of size or liquidity.

Table 13 presents 95% bootstrap confidence intervals for mean metrics, computed by resampling the 27 tickers with replacement (10,000 iterations).

Table 13: Bootstrap 95% confidence intervals for mean metrics ($n = 27$ tickers)

Strategy	Metric	Mean	CI Low	CI High
Buy & Hold	Return (%)	11.66	- 1.95	28.57
Buy & Hold	Sharpe	0.167	−0.011	0.371
Buy & Hold	Max DD (%)	−34.57	−37.80	−31.42
TA+CLT Fused	Return (%)	1.36	0.24	2.56
TA+CLT Fused	Sharpe	0.233	0.199	0.267
TA+CLT Fused	Max DD (%)	−18.09	−18.79	−17.47
TA Only	Return (%)	0.09	−0.76	0.95
TA Only	Sharpe	0.103	0.068	0.136
TA Only	Max DD (%)	−16.81	−17.31	−16.39

The Fused strategy’s Sharpe CI ([0.199, 0.267]) lies entirely above zero and entirely above TA Only’s upper bound (0.136), indicating that an investor applying the fused strategy across these banking stocks can expect reliably positive risk-adjusted returns. By contrast, Buy & Hold’s Sharpe CI ([−0.011, 0.371]) includes zero, offering no statistical guarantee of positive risk-adjusted performance. TA Only’s return CI ([−0.76, 0.95]) also includes zero, indicating that technical signals alone do not reliably overcome Vietnam’s transaction costs (0.15% commission plus 0.1% sell tax). The Fused return CI ([0.24, 2.56]) lies entirely above zero, confirming that the deep learning component adds sufficient predictive edge to cover trading frictions.

4.3.3. Robustness Analysis

Two robustness checks examine whether the results hold under alternative cost assumptions and strategy configurations.

Table 14 scales the commission rate per side across three scenarios while holding sell tax (0.1%) and T+2 settlement constraints unchanged.

Table 14: Transaction cost sensitivity (mean return, %)

Strategy	0.15% (base)	0.20%	0.25%
Buy & Hold	11.66	11.55	11.45
TA Only	0.09	0.00	−0.08
TA+CLT Fused	1.36	1.18	0.99

TA+CLT Fused maintains positive mean returns across all tested commission scenarios, retaining +0.99% even at the highest cost assumption (0.25% per side). TA Only becomes marginally unprofitable at 0.25% (−0.08%), indicating that its thinner edge is more sensitive to execution costs.

Table 15 varies the component weights (w_{AI} , w_{TA} , w_{fund}), stop-loss thresholds, and entry criteria to assess parameter sensitivity.

Table 15: Configuration sensitivity analysis (TA+CLT Fused strategy)

Configuration	Mean Ret.	Sharpe	vs. TA p	% better
Baseline ($w_{AI} = 0.45$)	+1.36%	0.233	0.073	55.6%
AI-heavy ($w_{AI} = 0.50$)	+0.88%	0.236	0.041	59.3%
TA-heavy ($w_{TA} = 0.40$)	+0.88%	0.171	0.036	59.3%
Risk-tight	+0.33%	0.197	0.075	63.0%
Defensive	−0.49%	0.105	0.585	55.6%

The configuration analysis reveals a clear trade-off between return and risk that maps to different investor profiles. The Baseline configuration ($w_{AI} = 0.45$) maximises mean return (+1.36%), while the Risk-tight profile achieves the highest ticker win rate (63.0%) at the cost of lower absolute return (+0.33%). The Defensive configuration (−0.49% return, $p = 0.585$) demonstrates that under-weighting the deep learning signal removes the strategy's edge entirely, confirming that the CLT component is the primary source of added value. No single configuration dominates across all metrics, indicating that

parameter selection involves an inherent trade-off between return potential, drawdown control, and statistical significance. In practice, the appropriate configuration depends on the investor's specific constraints and objectives.

5. Conclusion and Recommendations

5.1. Conclusion

The current investigation successfully engineered and validated a comprehensive advisory framework explicitly tailored for domestic banking equities. This architecture seamlessly merges a multi scale neural forecasting algorithm with a semantic routing extraction pipeline. Empirical testing utilized extensive historical market data spanning from January 2010 to March 2026 across 27 listed commercial institutions. All evaluations strictly incorporated local execution frictions to ensure absolute practical applicability.

Addressing the initial research inquiry regarding algorithmic efficacy reveals that the proposed neural predictor delivered robust directional accuracy. This topology consistently outperformed conventional algorithmic baselines on the tripartite classification problem. Managing the severe prevalence of neutral market conditions required specialized optimization functions, confirming the critical role of asymmetric loss penalties in financial machine learning (Lin et al., 2017). Attempting to incorporate quarterly fundamental ratios directly into the daily temporal sequence degraded forecasting precision substantially. This degradation cements the architectural decision to isolate accounting metrics within the retrieval module rather than embedding them inside the predictive engine.

Evaluating the qualitative component highlights substantial improvements in factual reliability. The partitioned semantic extraction infrastructure produced highly relevant documentary evidence, achieving exceptional accuracy in both routing operations and answer groundedness. Isolating distinct knowledge domains successfully mitigated the semantic interference typically observed in monolithic databases. This rigorous partition guarantees that synthesized advisory narratives remain strictly anchored to verifiable regulatory texts and fundamental disclosures.

Translating these quantitative outputs into portfolio execution yielded superior risk adjusted metrics. The integrated strategy significantly minimized maximum capital drawdowns compared to passive allocation frameworks. While the passive benchmark exhibited a superficially higher average return driven by a singular extreme outlier, the algorithmic approach provided broad based consistency across the entire banking sector. Modifying the component weights confirmed that prioritizing the deep learning signal constitutes the primary source of strategic advantage. The empirical evidence indicates the framework functions optimally as an advanced risk mitigation and decision support instrument rather than a pure excess return generator.

The aggregated findings fulfill all overarching research objectives while reinforcing established theoretical paradigms. The predictive viability of historical price geometries corroborates earlier propositions regarding informational inefficiencies within frontier exchanges (Brock et al., 1992; Fama, 1970; Lo et al., 2000; Park & Irwin, 2007). Furthermore, the demonstrated efficacy of compartmentalized extraction aligns perfectly with contemporary methodologies advocating for domain specific knowledge structuring to minimize generative fabrication (Gao et al., 2023).

Contextualizing this research within the broader academic discourse illuminates several core contributions. The predictive capacity mirrors established global benchmarks derived from developed markets (Fischer & Krauss, 2018; Kim & Kim, 2019), despite operating under stringent domestic liquidity constraints. While simulated reinforcement learning environments occasionally report higher absolute returns (Liu et al., 2021), the current methodology prioritizes structural realism by enforcing localized transaction execution costs. Finally, expanding foundational retrieval mechanics (Lewis et al., 2020) into specialized domestic financial dialects establishes a novel operational standard for natural language processing in low resource environments (Nguyen et al., 2024).

5.2. Recommendations

Translating the empirical findings into practical applications reveals distinct strategic advantages for two primary market participants. The actionable insights derived from this research target both retail equity traders and institutional wealth managers.

Retail participants navigating domestic banking equities must adopt the proposed integrated framework as a primary decision support mechanism to mitigate psychological trading biases. Utilizing the dual functionality of this architecture eliminates the traditional reliance on scattered news sources and emotional forecasting. Market participants can enforce rigorous portfolio discipline by executing the quantitative classification signals for precise market entries and stop loss triggers. Simultaneously, engaging the semantic extraction layer allows users to authenticate the underlying regulatory and financial rationale validating each quantitative alert. Adopting this comprehensive methodology equips ordinary individuals with a degree of data driven certainty previously confined to large institutional trading desks.

Institutional brokerages and asset management corporations should integrate the proposed pipeline as a foundational blueprint for advanced automated advisory infrastructures. Deploying this modular architecture facilitates the realisation of three core institutional objectives. Operational efficiency improves drastically because automating the synthesis of daily investment briefs occurs at a median latency of 2265 milliseconds per query, thereby alleviating the analytical burden placed on human research divisions. Furthermore, the inherent flexibility of the risk parameters demonstrates how institutions can customize portfolio allocations according to the precise risk return preferences of diverse clientele. Finally, the 87.3 percent groundedness metric guarantees that synthesized client communications remain fully auditable and heavily referenced, substantially lowering the legal vulnerabilities associated with algorithmic financial recommendations.

5.3. Limitations and Further Research

Certain boundaries define the current investigation and provide clear trajectories for subsequent scholarly work. The empirical validation strictly utilizes a specific universe of domestic commercial banks observed over a singular two year out of sample window. Restricting the analysis to a single economic phase naturally limits the capacity to project these findings onto entirely different corporate sectors or severe macroeconomic downturns. Future academic inquiries must validate this architecture across a much broader equity landscape using transfer learning methodologies. Subjecting the predictive

engine to extended historical evaluations spanning multiple market contractions and expansions will confirm its resilience under shifting structural realities.

Evaluating the quantitative outputs reveals moderate predictive strength rather than overwhelming directional certainty. While the deep learning module generates consistently positive risk adjusted metrics, the absolute portfolio returns remain suggestive rather than definitively superior to basic technical strategies. These performance parameters naturally trail the theoretical peaks reported by complex autonomous architectures operating within highly liquid Western exchanges (Liu et al., 2021). Overcoming this performance ceiling requires transitioning from static capital allocation rules toward dynamic position sizing governed by reinforcement learning algorithms. Furthermore, refining the neural topology through expanded temporal observation windows or multimodal data integration promises additional accuracy gains without improperly blending quarterly accounting metrics into the daily trading sequence.

Assessing the qualitative extraction component introduces another structural boundary. The current validation procedure depends on a manually constructed dataset comprising a highly specific number of test queries. Although meticulously stratified across distinct financial knowledge domains, this sample size restricts a comprehensive understanding of potential algorithmic hallucinations. Subsequent research endeavors must scale the evaluation infrastructure by integrating automated scoring frameworks and rigorous human oversight protocols similar to established generative testing standards (Friel et al., 2025). Expanding the testing parameters will expose systemic edge cases and refine the capacity of the advisory system to process highly ambiguous user requests.

Appendix: Sample Advisory Queries

Example 1: Single-stock advisory (CTG, 27/06/2025)

Q: Co phieu CTG co nen mua khong?

[Route: Fundamentals]

A: Ngay 27/06/2025, mo hinh CLT dua tren 43 chi tieu ky thuat (trend, momentum, volatility, volume) cho tin hieu MUA (score: 0.618, do tin cay CAO). Gia dong cua: 41,950 VND [1]. Khuyen nghi ky thuat: MUA voi cat lo -7%, chot loi +15%, nam giu toi da 30 ngay.

[1] CLT_Signal_CTG_2025-06-27

Example 2: Comparative signal (TCB vs VCB, 02/06/2025)

Q: So sanh TCB va VCB de dau tu ngan han, ma nao co tin hieu MUA?

[Route: Fundamentals | Decomposed: 2 sub-queries]

A: Ngay 02/06/2025, CLT cho TCB tin hieu MUA (score: 0.588, do tin cay TRUNG BINH), gia 30,850 VND [1]. VCB o trang thai GIU, khong co tin hieu MUA/ban [2]. Theo ket qua backtest (ky thuat + CLT), TCB co Sharpe 0.35, VCB 0.19 [3]. Tong hop: TCB phu hop hon cho ngan han theo tin hieu mo hinh.

[1] CLT_Signal_TCB_2025-06-02

[2] CLT_Signal_VCB_2025-06-02

[3] Fundamentals [01]

Example 3: Signal and risk parameters (MBB, 30/05/2025)

Q: MBB co tin hieu mua khong, cat lo chot loi nhu the nao?

[Route: Fundamentals]

A : Ngay 30/05/2025, CLT ghi nhan MBB o trang thai MUA (score: 0.582, do tin cay TRUNG BINH), gia 24,350 VND [1]. Tham so quan ly rui ro theo chien luoc: cat lo -7%, chot loi +15%, nam giu toi da 30 ngay. Nen dat lenh cat lo ngay khi vao lenh.

[1] CLT_Signal_MBB_2025-05-30

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